LaTeX with Overleaf + AI: After the session

Oct 3, 2026 · @Nitk Manoj

What we covered

You can now write, compile and share a LaTeX document entirely in Overleaf, with nothing installed.

1. Why LaTeX: you describe structure in plain text, and layout and numbering are automatic.
2. Overleaf basics: projects, the Code and Visual editors, maths, figures, tables and citations.
3. Going further: multi-file documents, your own commands, useful packages and beamer slides.
4. Diagrams as code: flowcharts, sequence diagrams and class diagrams that compile inside Overleaf.
5. AI in the workflow: generating tables and diagrams, fixing errors, and what to verify.
6. Hands-on: a one-page mini paper built from a blank project.

Session files

Two files come with this note: the slides as a PDF, and the demo project as `latex-session-demo.zip`.

To open the demo project in your own account:

1. Log in at overleaf.com and choose New project, then Upload project.
2. Select `latex-session-demo.zip`. Do not unzip it first.
3. Press Recompile, or Ctrl+Enter.
4. In `main.tex`, remove the `%` in front of one `\input` line at a time and recompile to see each topic appear.

The project is laid out the way a real paper or report should be:

```
main.tex            preamble, title, and one \input line per section
refs.bib            the reference list
sections/           one file per topic, 01-basics to 06-broken
figures/plot.png    the image used in the figure example
data/results.csv    a small table to convert with AI
prompts.txt         the AI prompts from the session
extras/slides.tex   a beamer slide deck
```

`06-broken.tex` contains three deliberate errors. Fixing them is practice task 3 below.

Cheat sheet

These commands cover most of what a paper or report needs.

To get	Type
Headings	`\section{Title}` and `\subsection{Title}`
Bold and italic	`\textbf{bold}` and `\textit{italic}`
Bullet list	`\begin{itemize} \item One \end{itemize}`
Numbered list	`\begin{enumerate} \item One \end{enumerate}`
Inline maths	`$E = mc^2$`
Numbered equation	`\begin{equation} ... \label{eq:one} \end{equation}`
Image	`\includegraphics[width=0.8\linewidth] {plot.png}` inside a `figure`
Table	`tabular` inside a `table`, with `\toprule`, `\midrule`, `\bottomrule` from booktabs
Cross-reference	`\label{fig:loss}` then `Figure~\ref{fig:loss}`
Citation	`\cite{key}`, with `\bibliography{refs}` at the end
Table of contents	`\tableofcontents`
Margins	`\usepackage[a4paper, margin=2.5cm] {geometry}`
New page	`\newpage`
Comment	`%` hides the rest of the line

The characters `# $ % & _ { }` need a backslash in front to print.

In Overleaf, Ctrl+Enter recompiles and Ctrl+/ comments or uncomments the selected lines. On a Mac, use Cmd.

Diagrams as code

Three packages draw diagrams from text inside Overleaf, with the default pdfLaTeX compiler and nothing to upload.

Diagram	Package	Key commands
Flowchart	`tikz`	`\node[style] (name) {text};` and `\draw (a) -- (b);`
Sequence diagram	`pgf-umlsd`	`\newthread`, `\newinst`, and the `call` environment
Class diagram	`pgf-umlcd`	the `class` environment, `\attribute`, `\operation`, `\inherit`

Working examples of all three are in `sections/05-diagrams.tex` of the demo project.

PlantUML and Mermaid do not compile reliably inside Overleaf. To use a PlantUML diagram, render it at plantuml.com, download the PNG, upload it to the project's file tree and insert it with `\includegraphics`.

Using AI

AI is fast at writing LaTeX and at explaining errors, and unreliable on facts, so you check everything it produces.

Prompts from the session, also in `prompts.txt`:

- **Table:** "Convert the data below into a LaTeX table using booktabs. Right-align the numeric columns, add a caption and \label{tab:results}. Return only the table environment."
- **Debug:** "My Overleaf project fails with the error below. Here are the 15 lines around it. Explain the cause in one sentence and give the smallest fix."
- **Convert:** "Rewrite this paragraph as LaTeX for the IEEEtran class. Keep my wording exactly; only add markup, maths mode and \cite placeholders."
- **Diagram:** "Write a pgf-umlsd sequence diagram: a user uploads a file, the server validates it, stores it and returns an ID. It must compile with pdfLaTeX on Overleaf. Return only the code."

Overleaf's own AI is switched on under Account Settings, AI features. The free plan allows 5 AI uses a day, and Error Assist is the one to spend them on.

Before you trust an AI answer:

- [ ] Every reference comes from Google Scholar, the DOI or the publisher, never from the chatbot's memory.
- [ ] Every equation is compared with its source. A wrong sign still compiles.
- [ ] Any package or command that is new to you is checked on CTAN.
- [ ] No confidential or unpublished material was pasted into a public AI tool.
- [ ] AI use is declared where your institute or the publisher asks for it.

Practice for the next week

Five short tasks turn today's session into a habit. Each takes 10 to 20 minutes.

- [ ] Finish the one-page mini paper from the hands-on: title, a list, one numbered equation, one table and one citation, each referred to with `\ref`, `\eqref` or `\cite`.
- [ ] Open a template from the Overleaf gallery that fits your work (IEEE paper, thesis or CV) and replace its sample text with your own.
- [ ] In the demo project, uncomment `\input{sections/06-broken}`, read the log and fix the three errors. Try without AI first, then compare with Error Assist.
- [ ] Draw one diagram from your own project as a TikZ flowchart or a pgf-umlsd sequence diagram.
- [ ] Take an existing Word report or assignment and move its first two pages into LaTeX.

Keep learning

Start with Overleaf's 30-minute guide, then use the others when you get stuck.

Resource	Use it for
Overleaf: Learn LaTeX in 30 minutes	A second pass over the basics
learnlatex.org	Short lessons with examples you can run online
TeX Stack Exchange	Searching an error message
CTAN	The manual of any package
Detexify	Finding a symbol by drawing it
LaTeX and Overleaf, Methods Bites	A beginner tutorial with a paper template

Reading list from the session:

- LianTze Lim, LaTeX: More Than Just Academic Papers and Theses
- Damodar Rajbhandari, A short introduction to LaTeX and its importance
- Gunnar Misund, The Perfect Thesis: generic thesis template
- D. S. Kulyabov, I. Tiddi and M. Jeusfeld, CEUR-WS paper template
- Overleaf blog, Overleaf AI now a part of Overleaf plans